Comptia N+ Assignment

[N+ - Network security, Maintenance and Troubleshooting procedures]

Wide Area Networks

Task:

❖ **1. Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.**

➢ **Aim**

- To understand and illustrate how a mobile phone connects to Instagram's servers through a home Wi-Fi network, Internet Service Provider (ISP), and Wide Area Network (WAN) devices.

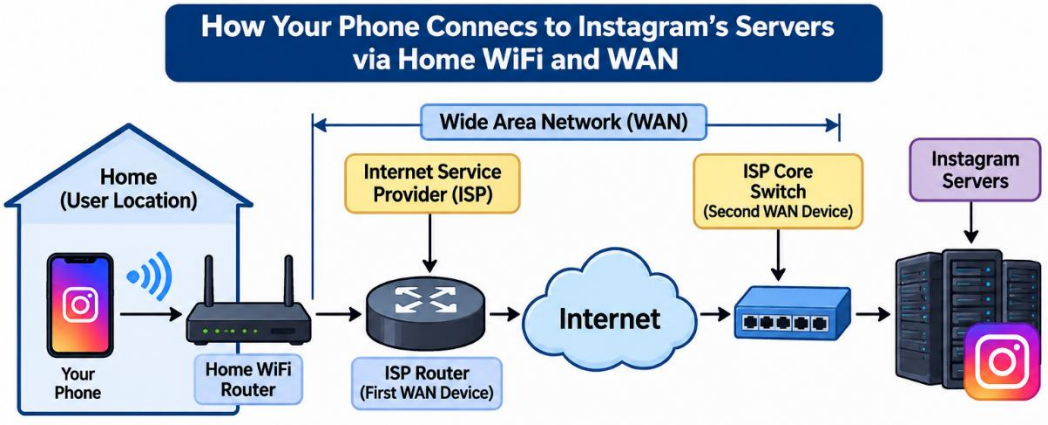
➢ **Introduction**

- A **Wide Area Network (WAN)** is a computer network that connects devices and smaller networks over large geographical areas. The Internet is the largest example of a WAN.
- When we open Instagram on our mobile phone using home Wi-Fi, our phone connects to the home Wi-Fi router. The router forwards the data

to the Internet Service Provider (ISP). The data then travels through the ISP's network and other WAN devices before reaching Instagram's servers.

- **Network Devices Used**
  1) **Mobile Phone:** Used to open Instagram and send or receive data.
  2) **Home Wi-Fi Router:** Connects the mobile phone to the home network and forwards data towards the ISP.
  3) **ISP Router:** Routes network traffic from the ISP's access network towards other networks.
  4) **ISP Core Switch:** Connects network links and helps forward traffic within the ISP's network.
  5) **Instagram Servers:** Receive requests and provide Instagram services, such as posts, messages, and images.

- **Network Diagram**
- The diagram illustrates the connection between a mobile phone and Instagram's servers through a home Wi-Fi network and WAN infrastructure.



- **Working of the Network**
  1) The mobile phone connects to the home Wi-Fi router through a wireless connection.
  2) The home Wi-Fi router forwards the data towards the Internet Service Provider (ISP).
  3) The ISP router routes the data through the ISP's network.
  4) The data travels through the Internet and other WAN infrastructure, including the ISP's core network devices.
  5) Finally, the request reaches Instagram's servers, which process it and send the required response.
  6) The response travels back through the network to the mobile phone, where Instagram displays the requested content.

- **Importance of WAN in This Communication**
  - WAN connects the user's home network to remote servers.
  - It allows communication over long geographical distances.
  - Routers help forward data between different networks.
  - It makes services such as Instagram, WhatsApp, and YouTube accessible from different locations.

- **Result**
- A network diagram was created to demonstrate how a mobile phone connects to Instagram's servers through a home Wi-Fi router, ISP network, and WAN devices.

❖ **2. Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).**

➢ **Aim**

- To study and compare three common WAN technologies—MPLS, Leased Line, and Broadband Internet—based on their speed, cost, features, and real-world applications.

➢ **Introduction**

- A **Wide Area Network (WAN)** is a network that connects computers, offices, and other networks over large geographical distances. Organizations use WAN technologies to communicate between different branches, access centralized servers, and share information.
- MPLS, Leased Line, and Broadband Internet are three commonly used WAN connectivity options. Each technology has different features, costs, and applications.

➢ **Comparison of Three WAN Technologies**

No.	WAN Technology	Speed	Cost	Real-World Usage
1	MPLS (Multiprotocol Label Switching)	High and predictable performance, depending on the	High cost compared to ordinary broadband.	A large organization such as a Zomato office may use an MPLS-based private WAN to

		service provider and network design.		connect multiple office branches and centralized systems.
2	Leased Line	Dedicated bandwidth with consistent and symmetrical speeds, depending on the selected plan.	High cost because it provides a dedicated connection.	A large service provider or organization operating important systems, such as an IRCTC data centre, could use leased-line connectivity for reliable communication between locations.
3	Broadband Internet	Variable speed depending on the service plan, network congestion, and technology used.	Low to moderate cost.	A person can use broadband at home to connect a mobile phone, laptop, or smart TV to the Internet.

- **Explanation of WAN Technologies**
- **1. MPLS (Multiprotocol Label Switching)**
  MPLS is a WAN technology that forwards network traffic using labels. It helps organizations manage communication between multiple locations and can support predictable network performance.
  Example: A company with offices in different cities can use an MPLS-based WAN to connect its branches.
- **2. Leased Line**

A leased line is a dedicated telecommunications connection between two locations. It provides consistent bandwidth and is commonly used by organizations that require reliable network connectivity.
Example: A large organization may use leased lines to connect its office with a data centre.

- **3. Broadband Internet**

  Broadband Internet provides high-speed Internet access through technologies such as fibre-optic cable, cable, or DSL. It is commonly used in homes and small businesses because it is generally more affordable.
  Example: A person uses a home broadband connection to access Instagram, YouTube, online banking, and other Internet services.

- **Main Differences**

  1) **MPLS:** Suitable for organizations that need managed communication between multiple branches.
  2) **Leased Line:** Suitable for organizations that require a dedicated and consistent connection between locations.
  3) **Broadband Internet:** Suitable for home users and small businesses that need affordable Internet access.

- **Result**
- Three WAN technologies—MPLS, Leased Line, and Broadband Internet—were studied and compared based on speed, cost, and real-world applications.

❖ **3. Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop. Focus on how the VPN creates a private tunnel over the public WAN.**

➢ **Aim**

- To understand how a Virtual Private Network (VPN) uses a Wide Area Network (WAN) to create a secure private tunnel over public Wi-Fi while accessing applications such as Spotify and YouTube.

➢ **Introduction**

- A **VPN (Virtual Private Network)** is a technology that creates a secure and encrypted connection between a user's device and a VPN server over a public network such as the Internet.
- A Wide Area Network (WAN) connects devices and networks over large geographical distances. The Internet is the largest example of a WAN.
- When a user connects to public Wi-Fi at a coffee shop, the network is shared with other users. A VPN helps protect the user's network traffic by encrypting it and sending it through a secure private tunnel over the public WAN.

➢ **How a VPN Creates a Private Tunnel**

➢ The VPN works through the following steps:

1) **Connect to Public Wi-Fi:** The user connects their mobile phone or laptop to the coffee shop's Wi-Fi network.

2) **Start the VPN:** The user opens a VPN application and connects to a VPN server.
3) **Create an Encrypted Tunnel:** The VPN establishes an encrypted connection between the user's device and the VPN server.
4) **Send Data Through the WAN:** The encrypted data travels through the public Internet (WAN). Other users on the public network cannot normally read the protected traffic.
5) **Access the Application:** The VPN server forwards the user's request to the required service, such as Spotify or YouTube.
6) **Receive the Response:** The response returns through the VPN connection and is decrypted by the user's device.

- **Example: Accessing Spotify or YouTube at a Coffee Shop**
- Suppose a person visits a coffee shop and connects their mobile phone to public Wi-Fi to listen to music on Spotify.
- Without a VPN, the user's traffic travels through the coffee shop's network and the Internet. HTTPS still encrypts the connection to Spotify, but other risks may remain on an untrusted network.
- With a VPN, the phone creates an encrypted tunnel to the VPN server. The traffic travels through the public WAN in encrypted form between the device and the VPN server. The VPN server then communicates with Spotify or YouTube.
- This provides an additional layer of protection for the user's network traffic while using public Wi-Fi.

- **Important Features of a VPN**
  1) **Encryption:** Protects data by converting it into an unreadable form for anyone who does not have the required keys.
  2) **Private Tunnel:** Creates a protected communication path over a public network.
  3) **Secure Remote Access:** Allows users to connect securely to remote networks and services.
  4) **Privacy:** Helps protect network traffic from being easily monitored by other users on the same public Wi-Fi network.

- **Result**
- The working of a VPN over a Wide Area Network was studied. It was understood how a VPN creates an encrypted private tunnel over the public Internet to provide an additional layer of security while accessing applications such as Spotify and YouTube.

❖ **4. Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.**

➢ **Aim**

- To recommend a suitable WAN technology for connecting Flipkart's warehouses in Ahmedabad, Mumbai, and Delhi and justify the selection with two important reasons.

➢ **Introduction**

- A **Wide Area Network (WAN)** connects computers, offices, and networks over large geographical distances. Businesses use WAN technologies to connect their different branches, warehouses, and data centres so that they can share information and communicate efficiently.
- Flipkart has warehouses in different cities, such as Ahmedabad, Mumbai, and Delhi. These warehouses need reliable network connectivity to exchange information about inventory, customer orders, shipments, and delivery operations.

➢ **Recommended WAN Technology: MPLS**

- **MPLS (Multiprotocol Label Switching)** is the recommended WAN technology for connecting Flipkart's warehouses in Ahmedabad, Mumbai, and Delhi.
- MPLS forwards network traffic using labels and helps organizations manage communication between multiple locations through a service provider's network.

- **Reasons for Selecting MPLS**
- **Reason 1: Reliable and Predictable Network Performance**
- MPLS can support predictable network performance and Quality of Service (QoS). This is useful for business applications that require reliable communication, such as inventory management, order processing, and warehouse coordination.

- **Reason 2: Easy Connectivity Between Multiple Locations**
- MPLS allows an organization to connect multiple branches and warehouses through a managed WAN. Flipkart can connect its Ahmedabad, Mumbai, and Delhi warehouses so that they can exchange information and access centralized business systems.

- **Example of Network Connectivity**
- Flipkart can connect its three warehouses through an MPLS-based WAN:
  - Ahmedabad Warehouse
  - Mumbai Warehouse
  - Delhi Warehouse
- All three locations can communicate through the service provider's MPLS network and share important business information.

- **Result**
- MPLS was selected as a suitable WAN technology for connecting Flipkart's warehouses in Ahmedabad, Mumbai, and Delhi because it supports managed connectivity, predictable network performance, and communication between multiple locations.